



RECOVERY RUNBOOK

VIRTUAL CONTROL
VMWARE CLOUD FOUNDATION SOLUTIONS

SDDC Manager UI Stuck "VMware Cloud Foundation is Initializing"

Diagnosing and repairing the vCenter SSH host-key mismatch that blocks PSC/SSO initialization after a vCenter redeploy, in a FIPS-enabled SDDC Manager (KB 316028).

SDDC MANAGER

SSH HOST KEY

PSC / SSO INIT

KB 316028

FIPS

VCF 9.0

VMware Cloud Foundation

SDDC Manager UI Stuck on "VMware Cloud Foundation is Initializing" — vCenter SSH Host-Key Repair

Prepared by: Virtual Control LLC **Date:** June 5, 2026 **Document Version:** 1.0 **Classification:** Internal **Platform:** VMware Cloud Foundation 9.0 — SDDC Manager (FIPS-enabled) **Primary reference:** Broadcom KB 316028

****Scope.**** Diagnosis and repair of the vCenter SSH host-key mismatch that blocks SDDC Manager PSC/SSO initialization — the UI stuck on "VMware Cloud Foundation is initializing" — after a management vCenter redeploy, on a FIPS-enabled SDDC Manager. Every command is shown as executed, with pass criteria.
****Companion docs.**** - [VCF-Operations-Integration-Trust-Repair-After-vCenter-Cert-Regen-2026-06-05.md] (./VCF-Operations-Integration-Trust-Repair-After-vCenter-Cert-Regen-2026-06-05.md) — VCF Operations vCenter/vSAN adapter repair for the same vCenter regeneration event

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1. Executive Summary

After the management vCenter was redeployed (which regenerates its SSH host key, machine SSL certificate, and SSO), the SDDC Manager UI would not load — it sat permanently on **"VMware Cloud Foundation is initializing..."**. SDDC Manager's REST API remained fully functional throughout (tokens, inventory, and tasks all responded), which localized the fault to the **UI initialization path**, not the core services.

The UI initialization runs a **PSC/SSO step** that retrieves the vCenter SAML **IdP metadata** via the local `commonsvcs` service. `commonsvcs` obtains that metadata by **opening an SSH session to the vCenter**. Because the vCenter's SSH host key had changed and SDDC Manager still trusted the old key in its `known_hosts` files, the SSH session was rejected, the metadata call returned HTTP 500, PSC init failed on every retry, and the UI never advanced past the initializing page.

The fix is Broadcom **KB 316028** — refresh the vCenter SSH host key across **all four** of SDDC Manager's `known_hosts` files. Because this SDDC Manager is **FIPS-enabled**, the KB's bash helper (`fix_known_hosts.sh`) is not usable; the FIPS-safe path is the `fixHostKeys.py` script or the equivalent manual `ssh-keyscan` refresh documented here. After refreshing the keys and restarting `sddc-manager-ui-app` , the metadata call returned HTTP 200 and the UI loaded normally.

Total time, single operator, no support engagement: approximately 45 minutes from log capture to a working login.

2. Was This an SSO Problem?

No — SSO was the victim, not the cause. The failure *surfaces* as an SSO/PSC initialization error ("Unable to retrieve iDP Metadata", "Primary psc init failed"), so it is easy to mis-diagnose as an SSO or certificate fault. The actual cause is one layer lower: **SSH host-key trust**. `commonsvcs` must SSH into vCenter to fetch the SSO IdP metadata; the SSH is what failed. SSO configuration on vCenter was healthy the entire time. Do not spend time on SSO/SAML or vCenter STS configuration for this symptom — go straight to the SSH host key.

3. Environment Reference

Item	Value
SDDC Manager	<code>sddc-manager</code> — 192.168.1.241
Management vCenter	<code>vcenter.lab.local</code> — 192.168.1.69
commonsvcs service user	<code>vcf_commonsvcs</code> (uid 994, home <code>/dev/null</code> , FIPS Java)
SSH login	<code>vcf user</code> , then <code>su -</code> to root — passwords vault refs <code><LAB_PW></code>
vCenter root SSH	<code>root@vcenter.lab.local</code> — password vault ref <code><vc-pw></code>
Four <code>known_hosts</code> files SDDC Manager uses	<code>/root/.ssh/known_hosts</code> , <code>/home/vcf/.ssh/known_hosts</code> , <code>/etc/vmware/vcf/commonsvcs/known_hosts</code> , <code>/opt/vmware/vcf/commonsvcs/defaults/hosts/known_hosts</code>

4. Symptoms

- Browser to `https://192.168.1.241/` shows **"VMware Cloud Foundation is initializing..."** and never advances.
- The SDDC Manager REST API still works (token acquire, `/v1/domains` , `/v1/vcenters` , `/v1/tasks` all respond).
- The vCenter VM web console (and, in this incident, the web console for **all** VMs) does not open — a *related but separate* vCenter console-proxy/cert symptom of the same redeploy; **not** fixed by this runbook.
- `/v1/vcenters` reports the management vCenter as **ACTIVATING** rather than **ACTIVE** .
- Periodic "Health-Check operation for SDDC" (SoS) tasks fail Pre-Validation with *"...the system could be in a transitional state."*

5. Root Cause Analysis

Root cause. The management vCenter was redeployed, which generated a new SSH host key. SDDC Manager still trusted the *previous* vCenter host key in all four of its `known_hosts` files. When `commonsvcs` attempted the

SSH session to vCenter to retrieve the SSO IdP metadata, host-key verification failed ("REMOTE HOST IDENTIFICATION HAS CHANGED"), the session was refused, and the metadata endpoint returned HTTP 500. The UI's PSC/SSO initialization loop therefore failed on every attempt, leaving the UI on the "initializing" page indefinitely.

Evidence chain (from `sddcManagerServer.log`):

```
axios GET http://127.0.0.1/commonsvcs/idpMetadata -> 500
axios.error.response.data: "Failed to establish SSH session to vcenter.lab.local"
VError: Unable to retrieve iDP Metadata: Request failed with status code 500
VError: Primary psc init failed and failover psc init also failed
PSC Initialization attempt "NNN" failed (looping)
```

Manual SSH confirmation (from the SDDC Manager root shell):

```
@@@ WARNING: REMOTE HOST IDENTIFICATION HAS CHANGED! @@@
Offending ECDSA key in /root/.ssh/known_hosts:2
Host key verification failed.
```

Why fixing only `/root/.ssh/known_hosts` was not enough. `commons` runs as `vcf_commons`, not root, and reads its own host-key store — chiefly `/etc/vmware/vcf/commons/known_hosts`. The stale `vcenter.lab.local` entries (an `ssh-rsa` and an `ecdsa-sha2-nistp256` line) lived there as well. All four files must be refreshed; KB 316028 does this, and the manual method below does the same.

6. Diagnosis — Commands

SSH to SDDC Manager as `vcf`, then `su -` to root.

6.1 Find the real initialization error (the startup-feedback server hides it):

```
grep -iE "psc init|idp metadata|status code 500|establish SSH" \
/var/log/vmware/vcf/sddc-manager-ui-app/sddcManagerServer.log | tail -n 30
```

The decisive line is `"Failed to establish SSH session to vcenter.lab.local"` against `http://127.0.0.1/commonsvcs/idpMetadata`.

6.2 Reproduce the failing call directly:

```
curl -sk -w "\nHTTP %{http_code}\n" http://127.0.0.1/commonsvcs/idpMetadata | tail -5
```

A **500** with "Failed to establish SSH session" confirms the SSH path is the blocker.

6.3 Confirm it is the host key (not credentials or a down sshd):

```
ssh -o ConnectTimeout=10 root@vcenter.lab.local "echo CONNECTED"
```

`WARNING: REMOTE HOST IDENTIFICATION HAS CHANGED` / `Host key verification failed` = host-key mismatch. (A password prompt that then fails = credential issue instead; `Connection refused` = sshd down on vCenter.)

6.4 Identify the service user and every host-key store:

```
ps -ef | grep -i commonsvc | grep -v grep      # runs as vcf_commonsvcs (FIPS Java)
find / -name known_hosts 2>/dev/null          # lists all four files
grep -n vcenter /etc/vmware/vcf/commonsvc/known_hosts # shows the stale rsa + ecdsa lines
```

7. Fix — Prescribed and FIPS-Safe Manual

FIPS caveat. This SDDC Manager is FIPS-enabled (`commonsvcs` runs with `-Dorg.bouncycastle.fips.approved_only=true`). KB 316028's bash helper `fix_known_hosts.sh` **does not work under FIPS**. Use the Python `fixHostKeys.py` method, or the manual `ssh-keyscan` method in 7.2 (also FIPS-safe — it is only file edits + key scanning).

Always take an SDDC Manager snapshot first (KB 316028 prerequisite). The vCenter UI is independent of this fault, so snapshot from vSphere Client or PowerCLI.

7.1 Prescribed method — `fixHostKeys.py` (KB 316028)

1. Download `fixHostKeys.py` from KB 316028 and place it in `/home/vcf/` on SDDC Manager.
2. As root: `sh python fixHostKeys.py --node vcenter.lab.local` - FIPS-safe; updates all four `known_hosts` files automatically. - Log: `/var/log/vmware/vcf/fixHostKeys.log` . - For all nodes of a type instead of one host: `python fixHostKeys.py --resourceType VCENTER` (also `NSX_T_MANAGER` , `ESXI` , `NSXT_EDGE`).

7.2 Manual method (used in this incident — no script download needed, FIPS-safe)

Run as root on SDDC Manager. This backs up each file, removes the stale `vcenter.lab.local` lines, and appends the current keys, **preserving each file's ownership and permissions** (the `grep -v ... > "$f"` truncates in place rather than recreating the inode):

```
ssh-keyscan -t rsa,ecdsa,ed25519 vcenter.lab.local 2>/dev/null > /tmp/vc_newkeys && \
for f in /root/.ssh/known_hosts /etc/vmware/vcf/commonsvc/known_hosts \
    /home/vcf/.ssh/known_hosts /opt/vmware/vcf/commonsvc/defaults/hosts/known_hosts; do \
    cp -a "$f" "$f.bak-2026-06-05" 2>/dev/null; \
    grep -v "vcenter.lab.local" "$f.bak-2026-06-05" 2>/dev/null > "$f"; \
    cat /tmp/vc_newkeys >> "$f"; \
done; echo DONE
```

When pasting into an interactive shell, paste it as a **single line** (strip the trailing backslashes/newlines) — a broken paste leaves the shell at a `>` continuation prompt; press `Ctrl-C` to recover.

7.3 Restart the UI service

```
systemctl restart sddc-manager-ui-app
```

The PSC/SSO init loop re-runs against the now-trusted SSH path.

8. Verification

8.1 The metadata call now succeeds:

```
curl -sk -w "\nHTTP %{http_code}\n" http://127.0.0.1/commonsvc/idpMetadata | tail -5
```

Expect **HTTP 200** returning the SAML EntityDescriptor XML (entityID `https://vcenter.lab.local/websso/SAML2/Metadata/vsphere.local`).

8.2 The UI app is serving (not the init placeholder):

```
curl -sk -o /dev/null -w "ui -> %{http_code}\n" https://127.0.0.1/ui/
```

A **302** redirect to the SSO login (rather than the static "initializing" page) means the real UI app is up. Hard-refresh `https://192.168.1.241/ui/` and log in.

8.3 No new PSC/SSH errors:

```
grep -iE "psc init|status code 500|establish SSH" \
/var/log/vmware/vcf/sddc-manager-ui-app/sddcManagerServer.log | tail -n 5
```

The errors stop accruing once the fix is in place.

9. In Plain English

SDDC Manager and vCenter recognize each other partly by a unique fingerprint — vCenter's **SSH host key**. When vCenter was rebuilt, it got a **brand-new fingerprint**, but SDDC Manager still had the **old one written down** in four places. So when SDDC Manager tried to phone vCenter to collect the single-sign-on settings it needs to start its dashboard, vCenter answered with the new fingerprint, SDDC Manager said *"that's not the fingerprint I have on file — I won't talk to you,"* and the dashboard sat forever on "initializing." The repair is simply to **erase the old fingerprint and write down the new one** in all four places, then restart the dashboard service. Nothing was actually wrong with single sign-on — it just couldn't be reached.

10. Lessons Learned

- **L1 — "initializing" + working API = UI init path, not core services.** If the REST API answers but the UI hangs on initializing, look at `sddcManagerServer.log` for the PSC/SSO init loop, not at the platform services.
- **L2 — The real error hides behind the startup-feedback server.** `tail` shows only "Incoming request for startup server." `grep` for `establish SSH` / `idp metadata` / `status code 500` to find the cause.
- **L3 — SSO/PSC init failure after a vCenter redeploy is usually an SSH host-key problem.** `commonsvcs` SSHes to vCenter to fetch IdP metadata. Confirm with a manual `ssh root@vcenter`.
- **L4 — There are FOUR known_hosts files; fixing root's alone is insufficient.** `commonsvcs` runs as `vcf_commonsvcs` and reads `/etc/vmware/vcf/commonsvcs/known_hosts`. KB 316028 (or the manual method) updates all four.
- **L5 — FIPS changes the tooling.** On a FIPS-enabled SDDC Manager, the bash `fix_known_hosts.sh` does not work; use `fixHostKeys.py` or the manual `ssh-keyscan` method.
- **L6 — Preserve ownership when editing service-owned files.** Use `grep -v ... > "$f"` (in-place truncate) rather than `ssh-keygen -R` so the service user retains read access; keep a per-file backup.
- **L7 — Verify with the exact failing call.** `curl http://127.0.0.1/commonsvc/idpMetadata` flipping 500 → 200 is the definitive success signal.

11. Command Appendix

```
# --- DIAGNOSIS (root on SDDC Manager) ---
grep -iE "psc init|idp metadata|status code 500|establish SSH" /var/log/vmware/vcf/sddc-manager-ui-app/sdd
cManagerServer.log | tail -n 30
curl -sk -w "%{http_code}\n" http://127.0.0.1/commonsvc/idpMetadata | tail -5
ssh -o ConnectTimeout=10 root@vcenter.lab.local "echo CONNECTED"
ps -ef | grep -i commonsvc | grep -v grep
find / -name known_hosts 2>/dev/null
grep -n vcenter /etc/vmware/vcf/commonsvc/known_hosts

# --- FIX (manual, FIPS-safe; single line in practice) ---
ssh-keyscan -t rsa,ecdsa,ed25519 vcenter.lab.local 2>/dev/null > /tmp/vc_newkeys && for f in /root/.ssh/kn
own_hosts /etc/vmware/vcf/commonsvc/known_hosts /home/vcf/.ssh/known_hosts /opt/vmware/vcf/commonsvc/def
aults/hosts/known_hosts; do cp -a "$f" "$f.bak-2026-06-05" 2>/dev/null; grep -v "vcenter.lab.local" "$f.ba
k-2026-06-05" 2>/dev/null > "$f"; cat /tmp/vc_newkeys >> "$f"; done; echo DONE

# --- FIX (prescribed, KB 316028; requires script download) ---
python fixHostKeys.py --node vcenter.lab.local

# --- RESTART + VERIFY ---
systemctl restart sddc-manager-ui-app
curl -sk -w "%{http_code}\n" http://127.0.0.1/commonsvc/idpMetadata | tail -5      # expect 200
curl -sk -o /dev/null -w "ui -> %{http_code}\n" https://127.0.0.1/ui/           # expect 302
```

12. Related Documents

- *VCF Operations Integration Trust Repair After vCenter Certificate Regeneration (2026-06-05)* — the companion fix for the VCF Operations vCenter/vSAN adapters after the same vCenter certificate regeneration.
- Broadcom KB 316028 — *How to update the SSH host keys on the SDDC Manager.*
- Broadcom KB 305970 / 417477 — *SDDC Manager UI fails to load with "VMware Cloud Foundation is initializing."*
- Broadcom KB 418872 — *SDDC Manager UI inaccessible following vCenter Machine SSL certificate replacement.*

13. Document Information

Field	Value
Title	SDDC Manager UI Stuck on "VMware Cloud Foundation is Initializing" — vCenter SSH Host-Key Repair
Document ID	VC-RUNBOOK-SDDCMGR-SSHHOSTKEY-20260605
Version	1.0
Issued	2026-06-05
Author	Virtual Control LLC
Classification	Internal
Platform	VMware Cloud Foundation 9.0 / SDDC Manager (FIPS-enabled)
Applies to	SDDC Manager UI stuck on "initializing" after a management vCenter redeploy / SSH host-key change
Primary reference	Broadcom KB 316028 (with KB 305970 / 417477 / 418872 as related)
Related Documents	VCF Operations Integration Trust Repair After vCenter Certificate Regeneration (2026-06-05)
Distribution	Lab notebook

Revision History

Version	Date	Notes
1.0	2026-06-05	Initial issue. Documents diagnosis (SSH-session failure behind the SSO/PSC init loop), root cause (stale vCenter host key across four known_hosts files), the FIPS caveat, the prescribed <code>fixHostKeys.py</code> and the manual <code>ssh-keyscan</code> fix, verification, and the SSO-is-the-victim clarification, with the full command set as executed.

Document End

See also: - *VCF Operations Integration Trust Repair After vCenter Certificate Regeneration (2026-06-05)* — companion fix for the VCF Operations vCenter/vSAN adapters after the same vCenter regeneration - Broadcom KB 316028 — *How to update the SSH host keys on the SDDC Manager*